

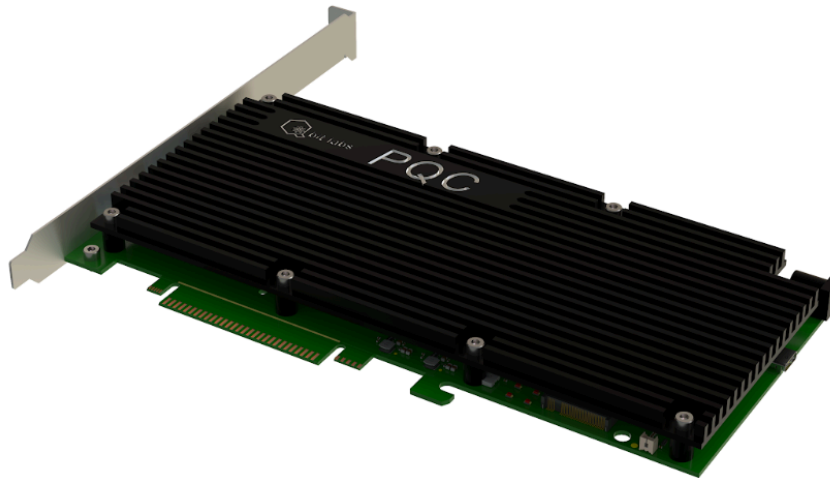
PQC Accelerator Endpoint (QPAE)

The QBIT PQC Accelerator Endpoint (QPAE) is a hardware-based solution designed to accelerate Post-Quantum Cryptography (PQC) by offloading complex cryptographic computations from the CPU. Built on a flexible FPGA architecture, it supports advanced quantum-resistant algorithms like ML-KEM and ML-DSA while also handling classical cryptography. By reducing processing bottlenecks and enabling easy updates for evolving standards, QPAE provides a scalable, vendor-independent, and future-proof alternative to traditional accelerators such as Intel QuickAssist Technology, ensuring improved performance, efficiency, and long-term security readiness.

Introduction

The QBIT PQC Accelerator Endpoint is a PQC endpoint interface designed to offload heavy cryptographic computation from the CPU. By offloading complex PQC operations, the QBIT PQC Accelerator Endpoint ensures that systems can adapt to emerging cryptographic algorithms, providing long-term security and enhancing system efficiency. The QPAE solution ensures that CPU systems can benefit from hardware-based acceleration of cryptographic operations without relying on integrated CPU functions or being limited by CPU capabilities. This acceleration will benefit both Intel, AMD and ARM systems.

QPAE supports the integration of PQC algorithms such as ML-KEM and ML-DSA, which are designed to withstand the potential threats posed by quantum computers. However, these algorithms are computationally intensive, necessitating substantial processing power and leading to performance bottlenecks, particularly in large-scale applications. Traditional cryptographic accelerators, such as Intel's QuickAssist Technology (QAT), offer limited flexibility to adapt to evolving PQC algorithms, resulting in higher maintenance costs and slower updates. A flexible, scalable solution, not dependent on a single CPU vendor, is required to accommodate the rapid evolution of PQC standards without compromising performance or security.



Architecture

The Qbit PQC Accelerator Endpoint (QPAE) is a next-generation, PCIe-based cryptographic interface designed to provide full hardware acceleration for both Post-Quantum Cryptography (PQC) and classical cryptographic standards. Engineered as a high-performance look-aside accelerator, the QPAE seamlessly integrates with host systems to offload computationally intensive operations, ensuring optimal system efficiency and security.

Designed for organisations navigating the critical transition to quantum-resistant security, the QPAE is built on a flexible FPGA architecture. This design provides unparalleled adaptability, allowing for the integration of custom and emerging algorithms, including the NIST-standardised Module-Lattice-Based Key Encapsulation Mechanism (ML-KEM) and Module-Lattice-Based Digital Signature Algorithm (ML-DSA). By offloading these heavy workloads from the host CPU, the QPAE ensures robust, future-proof protection without compromising system performance.

Why Choose Us?

-  Rapid integration, faster time-to-market
-  Clean, verified, documented RTL
-  Direct access to experts



Features

1. Advanced Algorithm Support

- Post-Quantum (PQC): Full hardware offload for ML-KEM (FIPS 203) and ML-DSA (FIPS 204) across all security levels (512/44 to 1024/87).
- Legacy/Classical: High-speed acceleration for RSA (up to 4096-bit) and ECC (P-256, P-384, P-521, X25519).
- Hybrid Security: Native support for X25519-MLKEM768, combining classical reliability with quantum resistance.

2. Seamless Ecosystem Integration

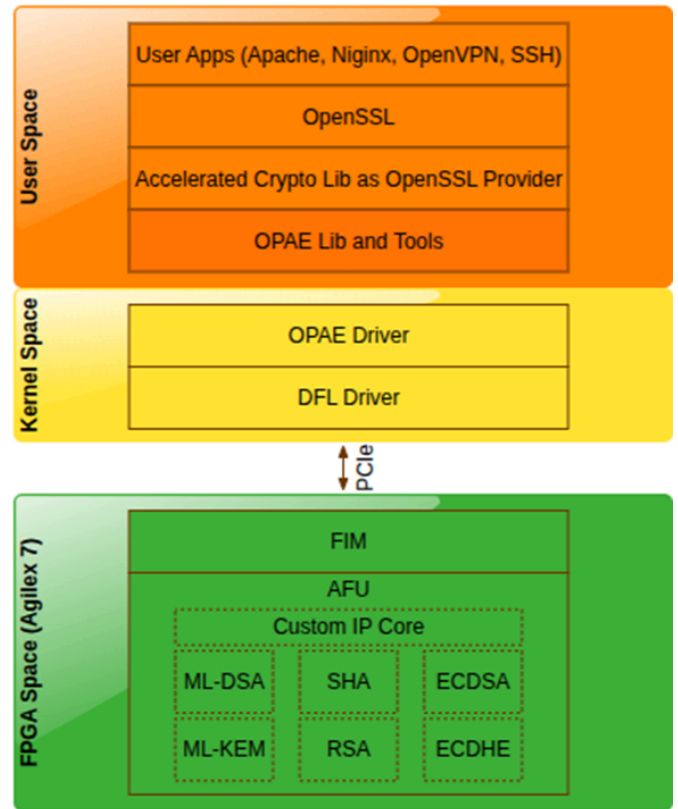
- Drop-in Ready: Transparent integration with OpenSSL 3.x, NGINX, and Apache via an optimized Provider.
- Modern Protocols: Built for TLS 1.3, ensuring future-proof secure communication with zero application code changes.
- OFS Framework: Leverages the Intel Open FPGA Stack for "plug-and-play" data center deployment.

3. Hardware-Rooted Security

- Entropy & Randomness: Integrated Physical TRNG (NIST SP 800-22/SP 800-90B) for superior key generation.
- Key Lifecycle: Secure on-chip key storage with hardened Import/Export/Rotation policies.
- Side-Channel Defense: Constant-time execution and RTL-level countermeasures against timing and power analysis attacks.

4. Operational Agility

- Future-Proofing: FPGA-based architecture allows for in-field updates to support emerging standards (e.g., Falcon, SPHINCS+) without hardware replacement.
- Efficiency: Offloads heavy lattice-based math from the Host CPU, reducing latency and increasing server throughput.



Specifications

Parameter	Details
Ports	• None
Host Interface	• PCIe interface for attachment to a host computer (PCIe Gen4x16).
DMA	• High Performance Low Latency Custom DMA
Key Encapsulation (Legacy)	• ECC: X25519, P-256, P-384, P-521 (ECDH / ECDSA)
Digital Signature (Legacy)	• RSA: KeyGen / Sign / Verify (2048, 3072, 4096-bit)
Key Encapsulation (PQC)	• ML-KEM-512, ML-KEM-768, ML-KEM-1024 (FIPS 203)
Digital Signature (PQC)	• ML-DSA-44, ML-DSA-65, ML-DSA-87 (FIPS 204)



Specifications Cont...

Parameter	Details
True Random Number Generator (TRNG)	NIST SP 800-22/SP 800-90B Compliant Entropy Source
Hybrid Schemes	Hardware Accelerated X25519_MLKEM768 Combiner
Reconfiguration	Partial Reconfiguration (PR) Support for Algo Updates
Cooling	Passive Heatsink (Server Airflow Required) or Active Fan Option
Operating Temp	0°C to 70°C (Commercial Grade)
Card Power	< 75W (Bus Powered) or < 150W (Aux 8-pin) depending on SKU
Card Dimensions	222 mm x106.7 mm

Deliverables

1. Host Software:

- Seamless plug-and-play integration with OpenSSL, NGINX, and Apache.
- Out-of-box driver support for Linux.

2. Hardware and FPGA:

- ½-height, ½-length PCIe card with Altera Agilex F FPGA PCIe Gen4 x16. (Form factors such as M.2, U.2, and E3.S may be offered)
- FPGA IP delivered in compiled code format with configuration scripts.

3. Management:

- Open FPGA Stack (OFS) features for easy FPGA management.

4. Documentation:

- Detailed datasheet, integration guide, and installation guide.

Licensing

- QBIT PQC Accelerator Endpoint

Get in touch with us

Delivering Advanced Solutions for
Next-Gen Networking



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